MOSTAFA ALI ZAKARIA

Software Engineer | Go Backend · Flutter · Applied AI

Cairo, Egypt | mostafa@studyzone.app

LinkedIn | GitHub | Available now for full-time remote or Cairo roles

PROFILE

Software engineer who built and scaled StudyZone to 359,000+ registered users (~165k MAU) across Android, iOS and web. Hands-on expertise across Go microservices, Flutter, PostgreSQL, distributed caching, payments, and production AI. Own architecture, implementation, automated testing, releases, and incident response; also engineered ShirtZone (shirtzone.app), an AI-powered custom apparel platform and commerce studio.

EXPERIENCE

StudyZone | Software Engineer and Founder Aug 2024 - Present

studyzone.app | App Store | Google Play

- Scaled platform organically to 359,000+ registered users (~165k MAU) and 1.6M+ question bank; owned architecture, code, on-call and 1,020+ merged PRs across mobile, web and backend.
- Published Flutter apps on Android, iOS and web with video learning, quizzes, subscriptions, chat and push notifications; structured features with Cubit/BLoC, Clean Architecture and offline persistence.
- Migrated backend from C#/.NET to Go (Fiber) with zero downtime; built 74 vertical-slice modules, REST APIs, WebSockets, Redis caching and 1,160+ versioned database migrations.
- Implemented payments and subscriptions with idempotent webhooks, provider-state reconciliation and server-side pricing to handle retries and eliminate duplicate fulfillments.
- Diagnosed production database contention and connection exhaustion; added singleflight cache-stampede protection, request deadlines and load shedding.
- Shipped production AI: multi-provider LLM orchestration (Gemini, OpenRouter), pgvector HNSW semantic search, AI study kits, automated evaluation fixtures, and daily spend governance.
- Built operational tooling including Flutter Web admin & teacher dashboards (admin.studyzone.app), YouTube curriculum ingestion, and automated media pipelines on Railway/Docker.

ADDITIONAL PRODUCT

ShirtZone | AI-Powered Custom Apparel Platform & Commerce Studio

shirtzone.app

- Architected and built an AI-powered custom apparel platform featuring "Stitch", an interactive studio that generates production-ready garment artwork from prompts with canvas editing and placement controls.
- Delivered the end-to-end commerce experience in Flutter (compiled to WebAssembly via skwasm and mobile) and Go: dynamic quote engine, multi-item cart, cash on delivery (COD) & card checkout, shipping, and virtual try-on, verified by 5,700+ automated tests.

TECHNICAL SKILLS

Backend and data: Go (Fiber), PostgreSQL (pgvector, TimescaleDB, SQL optimization), Redis, WebSockets, REST APIs, C#/.NET

Mobile and web: Flutter, Dart, Cubit/BLoC, Clean Architecture, responsive RTL/LTR UI, WebAssembly (skwasm), Firebase, FCM

AI and delivery: Multi-provider LLMs (Gemini, OpenRouter), embeddings & HNSW search, evaluations, Docker, Railway, GitHub Actions

EDUCATION

Bachelor’s Degree in Computer Science | New Cairo Academy, Egypt

Algorithmic problem solving: 1,000+ Codeforces problems solved.